Stage 4: Doctor / Patient Role Identification

Theory notes, alternative approaches and interview questions
Project: Doctor–Patient Conversation Audio -> SOAP Notes

What this stage does: turns anonymous labels (SPEAKER_00, SPEAKER_01) into roles (DOCTOR, PATIENT, OTHER). We implemented **Approach 1: rule-based linguistic cues**. This document also explains three alternatives (LLM, trained classifier, voice enrollment) so you can discuss what else could have been used and why.

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1. Why role identification matters for SOAP notes

SOAP section	Mostly comes from	Example
S - Subjective	PATIENT's own words	'My head has been hurting since yesterday'
O - Objective	DOCTOR's findings / measurements	'Let me check your blood pressure' -> 140/90
A - Assessment	DOCTOR's diagnosis	'It looks like a migraine'
P - Plan	DOCTOR's plan	'I will prescribe something... come back if worse'

If roles are swapped, the doctor's diagnosis becomes a 'patient complaint' and the patient's symptoms become 'clinical findings' - a dangerous error. That is why this stage outputs a **confidence** and flags weak decisions for human review.

2. Why diarization can't do it alone

- Diarization labels are **arbitrary per recording**: in clip 1 the doctor was SPEAKER_00, in clip 2 the doctor was again SPEAKER_00 but was the *second* person to speak and had a different voice.
- Diarization only knows voices are *different*, not *who* they belong to. A role must come from either **what** is said (approaches 1-3) or a **known voice** (approach 4).

3. The four possible approaches - overview

	1. Rules (ours)	2. LLM prompt	3. Trained classifier	4. Voice enrollment
Signal used	text: what is said	text: what is said	text: what is said	voice: how it sounds
Effort	low	low	high	medium
Accuracy	good on typical visits	very good	good (in-domain)	excellent (if enrolled)
Explainable	fully	partly	little	score only

4. Approach 1 (implemented): rule-based linguistic cues

Doctors and patients use language differently. We encode these differences as hand-written cues (regular expressions) and count them per speaker.

Doctor cues (+1 each per utterance)	Patient cues (+1 each per utterance)
Each question mark '?'	Own symptoms: 'I have been experiencing / having...'
Opening: 'what brings you', 'what seems to be the problem'	Body part: 'my head', 'my chest' ...
Duration: 'how long', 'when did it start'	Feelings: 'I feel', 'I'm worried', 'it hurts'
History: 'have you taken / had / noticed'	Own history: 'it started', 'I took', 'just paracetamol'

Doctor cues (+1 each per utterance)	Patient cues (+1 each per utterance)
Symptom: 'any allergies', 'do you smoke', 'on a scale'	About own condition: 'is it serious', 'should I worry', 'do I need'
Examination: 'let me check / listen'	
Plan: 'I'll prescribe / refer', 'blood test', 'come back'	
Assessment: 'it looks like', 'probably', 'diagnos...'	
First speaker: +0.5 (weak hint only)	

5. How our scoring works, step by step

```
for each speaker:
    doctor_points  = questions + doctor cues matched (+0.5 if first speaker)
    patient_points = patient cues matched
    score = (doctor_points - patient_points) / number_of_utterances

DOCTOR  = speaker with the highest score
PATIENT = among the rest, the one with the most patient_points
OTHER   = everyone else (relative, nurse, interpreter)

gap = score(top) - score(second)
confidence = HIGH if gap >= 1.0,  MEDIUM if gap >= 0.4,  else LOW  -> needs_review = true
```

- **Why divide by utterances?** Otherwise a speaker who simply talks more collects more points. Normalising compares *how* each person talks, not how much.
- **Why is 'first speaker' only +0.5?** Doctors often open the visit, but not always - in clip 2 the patient spoke first. It should only break near-ties.
- **Why count a cue once per utterance?** So a long sentence repeating a phrase can't dominate.
- **Evidence list**: every point is stored with the cue name and the sentence that triggered it - the decision is fully auditable.
- Input: outputs/<name>_aligned.json from Stage 3. Output: outputs/<name>_roles.json with roles, confidence, per-speaker evidence and a role-labelled transcript. No model, nothing installed.

6. Results on our test clips

Test	Difficulty	Result
Clip 1 (14 s)	Doctor speaks first, 0.5 s pauses	Correct. Scores +2.75 vs -1.00, HIGH confidence
Clip 2 (22.6 s)	Patient speaks first, voices swapped, patient asks 'Is it serious?', one-word 'Yes.', only 0.15 s pauses	Correct. Scores +2.00 vs -0.50, HIGH confidence
Small-talk only (unit test)	No medical cues at all	LOW confidence, flagged for review
3 speakers (unit test)	Doctor + patient + relative	DOCTOR / PATIENT / OTHER correct

In clip 2 the patient earned doctor points (a question, speaking first) but more patient points ('my head', 'I feel', 'since yesterday', 'is it serious'). Multiple weak cues combined beat any single

misleading cue - the core idea of evidence scoring.

7. Approach 2: Large Language Model (LLM)

- **How**: give the speaker-labelled transcript to an LLM with a prompt such as: 'Which speaker is the clinician, which is the patient, are there others? Reply as JSON with a reason and confidence.'
- **Why it works**: LLMs understand context and intent - e.g. that 'Is it serious?' is a worried patient and 'Have you had any nausea?' is history-taking - without hand-written rules.
- **Options**: a local open model (Llama, Qwen, Mistral via Ollama/llama.cpp; 2-5 GB) keeps data on device; a cloud API (Claude, GPT) is stronger but needs a healthcare data agreement (e.g. a HIPAA BAA).
- **Pros**: high accuracy, handles unusual phrasing, other languages, 3+ speakers, almost no code.
- **Cons**: privacy, cost/latency, possible hallucinated reasoning, less deterministic; must validate the JSON output and still keep a confidence / review step.

8. Approach 3: Trained text classifier

- **How**: fine-tune a small transformer (e.g. BERT, DistilBERT, or a clinical model like Bio_ClinicalBERT) to classify each utterance as doctor or patient. Then each speaker's utterance probabilities are averaged (soft voting) to decide the speaker's role.
- **Data**: public doctor-patient dialogue datasets with role labels: MTS-Dialog, ACI-Bench, PriMock57, MedDialog.
- **Pros**: learns cues automatically, fast and local at inference, measurable accuracy, no prompt sensitivity.
- **Cons**: needs labelled data and training; domain shift (different accents, specialties, ASR errors) lowers accuracy; less explainable than rules (can add SHAP/attention for insight).

9. Approach 4: Voice enrollment (speaker verification)

- **How**: record each doctor once (~30 s). pyannote's embedding model (the same WeSpeaker ResNet from Stage 2) turns it into a voiceprint vector. For each new consultation, compare every speaker's voiceprint to the enrolled doctors using **cosine similarity**; above a threshold = that doctor.
- Everyone not matching an enrolled clinician is PATIENT or OTHER (text cues can separate those two).
- **Pros**: most reliable in real deployments (same doctor every day), independent of what is said, works even for very short visits, and identifies *which* doctor.
- **Cons**: needs an enrollment step and threshold tuning; voiceprints are **biometric data** (GDPR special category) - consent and secure storage required; microphone changes and illness affect the voice.

10. Comparison table and hybrid design

	1. Rules (ours)	2. LLM	3. Classifier	4. Voice enrollment
Needs	nothing	LLM (local or API)	labelled data + training	doctor voice sample

	1. Rules (ours)	2. LLM	3. Classifier	4. Voice enrollment
Accuracy	good on typical visits	very high	high (in-domain)	very high
Explainability	full (evidence list)	reason text	low	similarity score
Privacy	local	local or cloud	local	local, but biometric
Speed / cost	instant, free	seconds, GPU or $	fast after training	fast
Fails when	unusual phrasing, small talk	prompt/model issues	domain shift	no enrollment, new doctor

Production design (hybrid): use voice enrollment when the doctor is enrolled; otherwise use the text method (rules or LLM); if methods disagree or confidence is low, ask a human to confirm in the UI. Rules remain useful as a cheap, explainable baseline and as a sanity check on the LLM.

11. Why we chose Approach 1

- Learning goal: every decision is visible and understandable.
- No new dependency, no model download, no data leaves the machine.
- Establishes a **baseline**: any fancier method must beat it to justify its cost.
- Its confidence + evidence output already defines the interface a stronger method would plug into.

12. Evaluation

- **Speaker-level accuracy**: % of recordings where every speaker gets the right role. The key metric.
- **Utterance-level accuracy / F1**: % of utterances with the right role (captures diarization + alignment errors too).
- **Calibration**: when confidence is HIGH, how often is it actually right? Thresholds (1.0 / 0.4) should be tuned so HIGH is almost never wrong.
- **Review rate**: % flagged LOW. A trade-off between safety and human workload.
- Use labelled datasets (e.g. PriMock57 has real mock consultations with audio) and compare all approaches.

13. Limitations of our rule-based approach

- English-only, hand-written patterns; misses paraphrases ('What's troubling you?').
- Short consultations or small talk give few cues -> LOW confidence (by design, flagged).
- Errors from earlier stages pass through: if diarization merges two people, rules can't fix it.
- Patients who ask many questions, or doctors who talk about their own experience, can mislead the score.
- Separating PATIENT from a talkative relative is hard (both use patient-like language).

14. Interview questions and answers

Q1. Why do you need doctor/patient identification after diarization?

Diarization gives arbitrary labels per recording. SOAP notes need to know whose words are symptoms (patient, Subjective) and whose are findings and plans (doctor, Objective/Assessment/Plan).

Q2. How does your rule-based method work?

It counts doctor cues (questions, history-taking, examination, prescribing, assessment) and patient cues (own symptoms, 'my ...', feelings, own history, worry) per speaker, normalises by number of utterances, and picks the highest net score as DOCTOR. Confidence is based on the gap between the top two scores.

Q3. Why normalise by utterances?

So the decision reflects how each person talks, not how much. Patients often talk more; raw counts would bias toward whoever speaks most.

Q4. Isn't 'the first speaker is the doctor' good enough?

No - our second clip had the patient speak first. We use it only as a +0.5 tie-breaker. Combining many weak cues is more robust than any single rule.

Q5. How do you handle uncertainty?

A confidence level from the score gap; LOW sets needs_review=true so a human confirms before the roles feed into the SOAP note. Every point is stored with its triggering sentence for audit.

Q6. How do you handle three or more speakers?

Highest score is DOCTOR, the remaining speaker with most patient cues is PATIENT, the rest are OTHER (relative, nurse, interpreter).

Q7. What alternatives did you consider?

An LLM prompt, a fine-tuned BERT utterance classifier with speaker-level voting, and voice enrollment with speaker embeddings. I started with rules as an explainable, private, zero-dependency baseline.

Q8. When would you use an LLM instead?

When phrasing varies a lot, multiple languages are involved, or rules give too many LOW-confidence cases. Use a local model or a cloud API under a BAA, require JSON output with a reason, and keep the review step.

Q9. How would you build the classifier approach?

Fine-tune a BERT-family model on role-labelled dialogue utterances (MTS-Dialog, ACI-Bench, PriMock57), predict per utterance, average probabilities per speaker, assign the role with the highest mean, and evaluate on held-out consultations.

Q10. Explain voice enrollment.

Record the doctor once, compute a speaker embedding, and compare each diarized speaker's embedding with cosine similarity; above a tuned threshold means it is that doctor. It's the most reliable in a clinic but involves biometric data, so consent and secure storage are needed.

Q11. What is cosine similarity?

The cosine of the angle between two vectors: dot(a, b) / (|a| |b|). 1 = same direction (very similar voices), around 0 = unrelated. Used to compare speaker embeddings.

Q12. What would a production system look like?

Hybrid: voice enrollment for known clinicians, text-based method as fallback, agreement checks between methods, and human confirmation in the UI when confidence is low.

Q13. How would you evaluate role identification?

Speaker-level accuracy per recording, utterance-level F1, calibration of confidence levels, and review rate, on a labelled set of consultations.

Q14. What are the risks of getting roles wrong?

Patient symptoms recorded as clinical findings or the doctor's diagnosis as a complaint - a patient-safety issue. Hence confidence, auditability and human review.

Q15. Why regular expressions and not just keyword lists?

Regexes match patterns with variation ('I have been experiencing', 'I've been having') and word boundaries (\bmy head\b) to avoid accidental matches inside other words.

Q16. How could rules and an LLM be combined?

Run rules first (cheap); call the LLM only for LOW-confidence cases; or use rules as a sanity check and flag disagreements.